



Nivell 1

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

Exercici 1

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

Lo primero que hice fue crear una tabla unificando los usuarios americanos y los usuarios europeos en una misma tabla.

```
162 -- crearemos una tabla nueva con todos los usuarios independientemente de que continente son --
163
164 • CREATE TABLE data_users AS
165 SELECT id, name, surname, phone, email, birth_date, 'America' AS continent, country, city, postal_code, address FROM american_users
166 UNION ALL
167 SELECT id, name, surname, phone, email, birth_date, 'Europe', country, city, postal_code, address FROM european_users;
168
169 • select * from data_users;
```

Result Grid										
Filter Rows:										
Export:										
Wrap Cell Content:										
Fetch rows:										
id	name	surname	phone	email	birth_date	continent	country	city	postal_code	address
22	Alistair	Holmes	1-265-323-0812	donec.tempor.est@protonmail.com	1990-11-05	America	United States	Dallas	75201	Ap #794-4229 Ante Rd.
23	Kelsie	Bass	1-837-832-5631	consequat@google.ca	1990-04-02	America	United States	San Antonio	78201	407-7562 A
24	Nolan	Cash	(273) 334-3785	nam@hotmail.com	1994-09-09	America	United States	Chicago	60601	501-2733 Luctus. Rd.
25	Wanda	Campbell	(702) 823-5535	sagittis@google.couk	1999-05-31	America	United States	Phoenix	85001	Ap #337-8747 Auctor. Ave
26	Aquila	Strickland	1-246-231-5495	enim.sit@icloud.com	1982-09-28	America	United States	Philadelphia	19101	Ap #260-4612 Massa Road

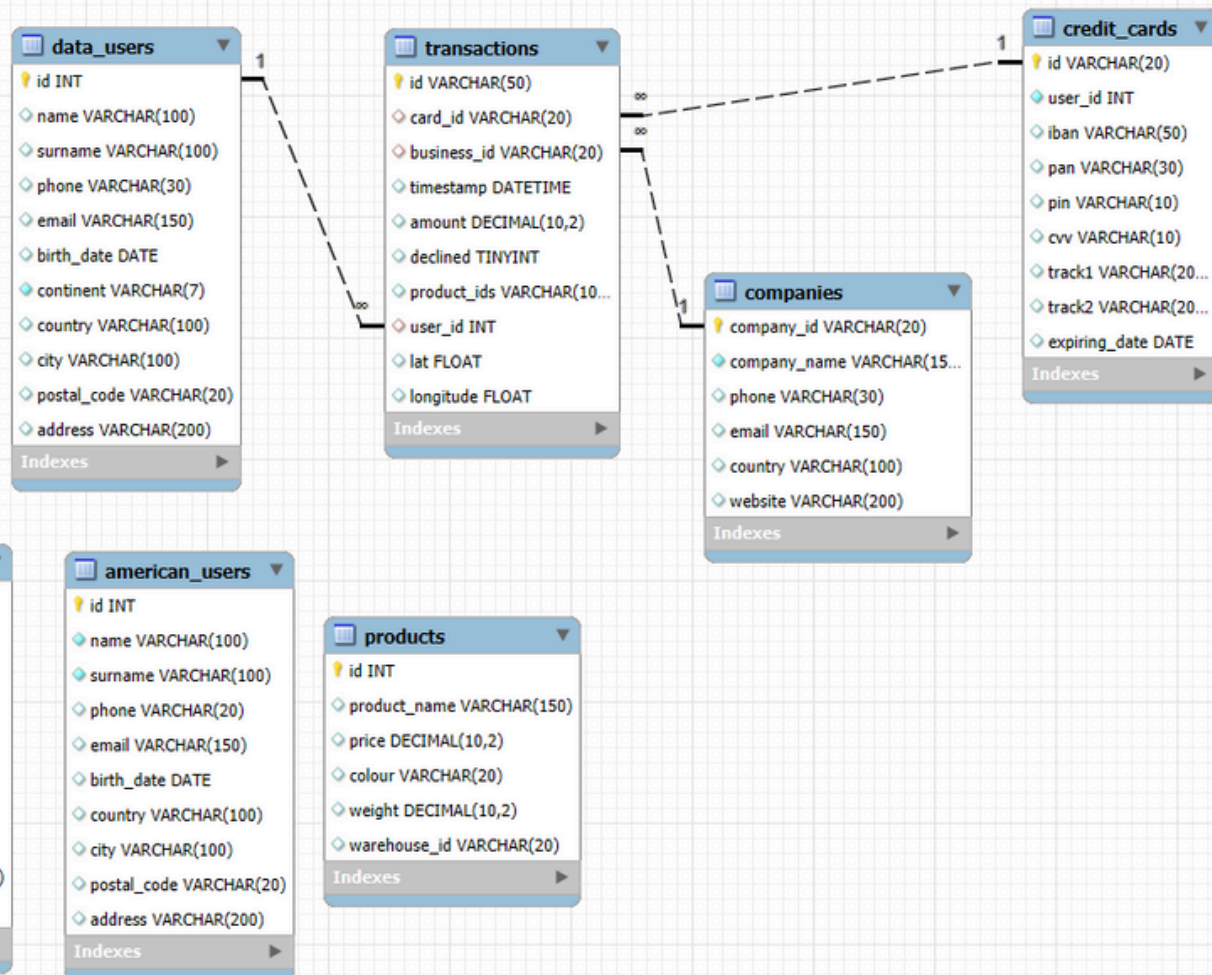
data_users 19 x

Output

Action Output

#	Time	Action	Message
7	13:47:54	SELECT id, COUNT(*) AS total FROM (SELECT id FROM american_users UNION ALL SELECT id FROM european_us...	0 row(s) returned
8	13:49:28	CREATE TABLE data_users AS SELECT id, name, surname, phone, email, birth_date, 'America' AS continent, country, city, pos...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
9	13:49:32	select * from data_users	5000 row(s) returned

Para que el modelo quede de la siguiente manera:



Luego resolví el ejercicio 1:

```

197 -- NIVEL 1 --
198 -- EJERCICIO 1 ---
199 -- Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules. --
200
201 • SELECT user_id, COUNT(user_id) AS "cantidad de transacciones", name, surname FROM transactions
202 JOIN data_users ON user_id = data_users.id
203 WHERE declined = 0
204 GROUP BY user_id
205 HAVING COUNT(user_id) > 80
206

```

Result Grid				
user_id	cantidad de transacciones	name	surname	
185	107	Molly	Gilliam	
289	91	Dxwgl	Hwvru	
318	86	Bnyr	Astuw	

#	Time	Action	Message	Duration / Fetch
24	10:56:53	select * from data_users LIMIT 0, 10	10 row(s) returned	0.000 sec / 0.000 sec
25	10:58:06	select user_id, count(user_id) AS "cantidad de transacciones", name, surname from transactions join data_users on ...	3 row(s) returned	0.437 sec / 0.000 sec
26	10:58:26	SELECT user_id, COUNT(user_id) AS "cantidad de transacciones", name, surname FROM transactions JOIN data_...	3 row(s) returned	0.391 sec / 0.000 sec

Exercici 2

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
207 -- EJERCICIO 2 --
208 -- Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.--
209
210 SELECT iban, AVG(amount), company_name FROM transactions
211 JOIN companies ON company_id = business_id
212 JOIN credit_cards ON credit_cards.user_id = transactions.user_id
213 WHERE company_name = "Donec Ltd" AND declined = 0
214 GROUP BY iban
215
```

Result Grid

iban	AVG(amount)	company_name
XX82380423389017414905481	9.240000	Donec Ltd
XX19575492990884857762085	348.800000	Donec Ltd
XX243111635090228872502782	288.770000	Donec Ltd
XX899873030823133018035771	325.410000	Donec Ltd
XX748890729057195711766071	607.290000	Donec Ltd
XX61501282784278671221718	514.330000	Donec Ltd

Result 6

Output

Action Output

#	Time	Action	Message	Duration / Fetch
63	12:29:20	select * from transactions where business_id = b'2242'	449 row(s) returned	0.000 sec / 0.000 sec
64	12:29:54	select iban, avg(amount) from transactions join companies on company_id = business_id join credit_cards on credit_...	370 row(s) returned	0.828 sec / 0.000 sec
65	12:31:21	SELECT iban, AVG(amount), company_name FROM transactions JOIN companies ON company_id = business_id J...	370 row(s) returned	1.031 sec / 0.000 sec



Nivell 2

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en *si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu*. Partint d'aquesta taula respon:

Exercici 1

Quantes targetes estan actives?

```

230 • CREATE TABLE credit_card_state AS
231 SELECT subquery.card_id AS card_id, CASE WHEN SUM(subquery.declined) = 3 THEN 'inactivo'
232 ELSE 'activo'
233 END AS state
234 FROM (SELECT card_id, timestamp, declined, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS rowi
235 FROM transactions
236 ) AS subquery
237 WHERE subquery.rowi <= 3
238 GROUP BY subquery.card_id;
239
240 • select * from credit_card_state;
241

```

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:

card_id	state
CcS-4857	activo
CcS-4858	activo
CcS-4859	activo
CcS-4860	activo
CcS-4861	activo
CcS-4862	activo
CcS-4863	activo

credit_card_state 21 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 46	11:07:04	drop table credit_card_state	0 row(s) affected	0.032 sec
✓ 47	11:07:17	CREATE TABLE credit_card_state AS SELECT subquery.card_id AS card_id, CASE WHEN SUM(subquery.decline...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	0.594 sec
✓ 48	11:07:22	select * from credit_card_state	5000 row(s) returned	0.000 sec / 0.016 sec

```

240 • select * from credit_card_state;
241
242 -- Cuantes tarjetas estan activas? --
243 • SELECT COUNT(state) AS tarjetas_activas from credit_card_state
244 group by state
245 having state = 'activo';
246
247 -- NIVEL 3 --

```

Result Grid Filter Rows: Export: Wrap Cell Content:

tarjetas_activas
4995



Nivell 3

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

Creo una tabla intermedia entre products y transactions con dos columnas, transaction_id y product_id, e incluí la columna con las transacciones declined

```
252
253 • ○ CREATE TABLE IF NOT EXISTS transaction_product (
254     transaction_id VARCHAR(50) NOT NULL,
255     product_id INT NOT NULL,
256     declined TINYINT NOT NULL,
257     PRIMARY KEY (transaction_id, product_id)
258 );
```

Le inserto los datos de la tabla transactions, separando cada transaccion con cada producto, haciendo una relacion de n a 1 con la tabla transactions media el transaction_id y de n a 1 con la tabla products mediante product_id


```

263 • INSERT INTO transaction_product (transaction_id, product_id, declined)
264 SELECT
265     transactions.id AS transaction_id,
266     CAST(lista_ids.product_id AS UNSIGNED) AS product_id, -- cambia de texto a numero entero positivo --
267     transactions.declined AS declined
268 FROM transactions
269 CROSS JOIN JSON_TABLE(
270     CONCAT('["', REPLACE(transactions.product_ids, ',', "',''), '"]'), -- json table para cada * incluir $
271     '$[*]' COLUMNS (product_id VARCHAR(20) PATH '$')
272 ) AS lista_ids
273 WHERE transactions.product_ids IS NOT NULL
274 AND transactions.product_ids <> '';
275
276
277 • SELECT * FROM transaction_product;

```

transaction_id	product_id	declined
00043A49-2949-494B-A5DD-A5BAE38819DD	16	0
00043A49-2949-494B-A5DD-A5BAE38819DD	26	0
00043A49-2949-494B-A5DD-A5BAE38819DD	87	0
00043A49-2949-494B-A5DD-A5BAE38819DD	97	0
000447FE-8650-4DCF-8SDE-C7ED0EE1CAAD	66	0
000447FE-8650-4DCF-8SDE-C7ED0EE1CAAD	69	0

#	Time	Action	Message	Duration / Fetch
4	11:33:14	CREATE TABLE IF NOT EXISTS transaction_product (transaction_id VARCHAR(50) NOT NULL, product_id IN...	0 row(s) affected	0.031 sec
5	11:36:10	INSERT INTO transaction_product (transaction_id, product_id, declined) SELECT transactions.id AS transaction...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	10.047 sec
6	11:37:34	SELECT * FROM transaction_product	253391 row(s) returned	0.000 sec / 0.203 sec

Creo las FK:

```

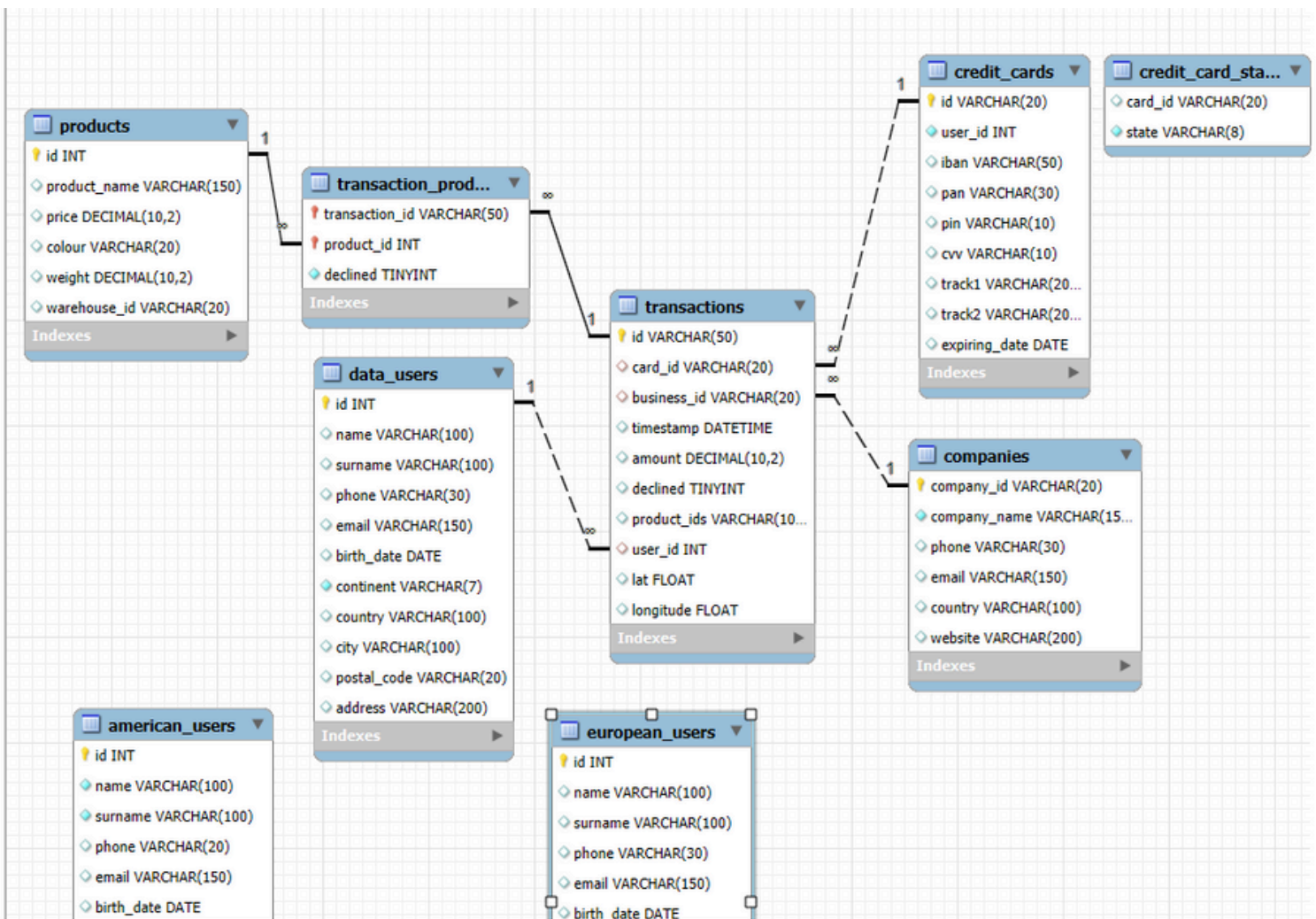
276 -- FK products.id
277 • ALTER TABLE transaction_product
278 ADD CONSTRAINT fk_transaction_product_products
279 FOREIGN KEY (product_id) REFERENCES products(id)
280 ON DELETE RESTRICT
281 ON UPDATE CASCADE;
282
283 -- FKtransactions.id
284 • ALTER TABLE transaction_product
285 ADD CONSTRAINT fk_transaction_product_transactions
286 FOREIGN KEY (transaction_id) REFERENCES transactions(id)
287 ON DELETE RESTRICT
288 ON UPDATE CASCADE;
289
290

```

transaction_id	product_id
001A60EA-DC9C-4E5A-9460-6628B100E7E1	1
0032F08B-8BE6-4AA5-B5EE-EEAD533C0C48	1
00342381-503D-422D-85AB-F2D4FFAAD4C7	1
004C0A80-E537-46D8-8E44-343D2176DF15	1
004D1D85-B2CB-4460-98B6-31C42CA96E5F	1

#	Time	Action	Message	Duration / Fetch
27	11:24:30	ALTER TABLE transaction_product ADD CONSTRAINT fk_transaction_product_products FOREIGN KEY (product...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	5.203 sec
28	11:24:39	ALTER TABLE transaction_product ADD CONSTRAINT fk_transaction_product_transactions FOREIGN KEY (trans...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	6.594 sec
29	11:26:38	SELECT * FROM transaction_product	253391 row(s) returned	0.000 sec / 0.187 sec

Para que quede el modelo de la siguiente manera:



Exercici 1

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

```
296 • SELECT products.id AS product_id,
297     products.product_name AS product_name,
298     COUNT(transaction_product.transaction_id) AS veces_vendido
299 FROM products
300 JOIN transaction_product ON transaction_product.product_id = products.id
301 WHERE declined = 0
302 GROUP BY products.id,
303          products.product_name
304 ORDER BY veces_vendido DESC, products.id;
305
306
```

Result Grid

Filter Rows:

Exports: Wrap Cell Content:

	product_id	product_name	veces_vendido
▶	52	riverlands the duel	2642
	29	Tully maester Tary	2627
	21	duel Direwolf	2603
	16	the duel warden	2602
	33	duel warden	2593
	87	sith Jade	2591
	...	...	...

Result 2 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	11:38:26	ALTER TABLE transaction_product ADD CONSTRAINT fk_transaction_product_products FOREIGN KEY (product...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	5.063 sec
8	11:38:34	ALTER TABLE transaction_product ADD CONSTRAINT fk_transaction_product_transactions FOREIGN KEY (trans...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	5.500 sec
9	11:38:45	SELECT products.id AS product_id, products.product_name AS product_name, COUNT(transaction_product.trans...	100 row(s) returned	1.015 sec / 0.000 sec